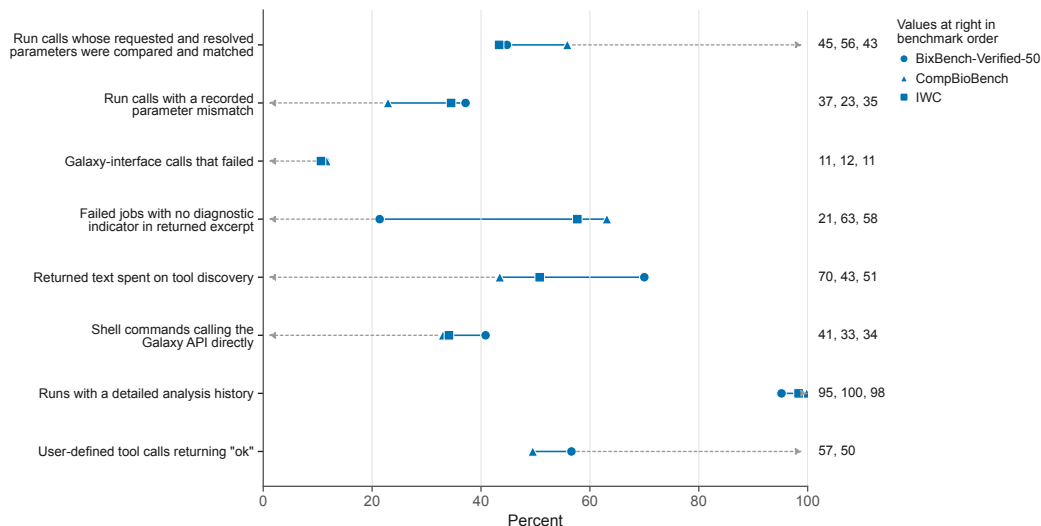


a Baseline measures (Galaxy condition; arrows: direction of improvement)



b Requirements motivated by the audit, their baseline measure and the intervention needed to validate them

Requirement	Baseline or supporting evidence	Intervention to test on held-out tasks
Bind parameters strictly and return what will run	Run calls with a recorded parameter mismatch	Reject unbound keys; return the resolved state before submission
Explain every failure in a structured form	Failed jobs with no diagnostic indicator in returned excerpt	Return phase, message and excerpt for every failed job
Expose versions and output semantics as tool metadata	No systematic baseline; five mechanism case studies	Add versions and per-output definitions to search and inspect
Make discovery compact and context-free	Returned text spent on discovery	Ranked tool cards; schemas without a history
Cover the analysis life cycle in the agent interface	Shell commands calling the Galaxy API directly	History copy, full download and resumable waits in the interface
Attest where computation ran and isolate runs	History coverage and location-uncertain cases (Fig. 5)	Per-run accounts; execution-location record in the history

None of these interventions was tested in this study. Measures describe archived adapter deployments on one server, July to September 2026.